

Project - || Presentation

Topic – Cloud-Based E-Commerce

Presented by:

Shantanu Halder - 27600122038

Amit Kumar - 27600122097

Saurav Suman - 27600122121

Dharmendra Kumar - 27600122187

Anjali Kumari - 27600122200

Supervisor: Mrs. Rituparna Dey

Department of Computer Science & Engineering



Budge Budge Institute of Technology, Kolkata

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❖ OBJECTIVE

- To build a secure and easy-to-use cloud-based e-commerce website.
- To allow users to browse, search, and buy products from anywhere.
- To ensure safe login, secure payments, and fast checkout.
- To help admins manage products, orders, and customers efficiently.
- To provide a smooth shopping experience using cloud technology.
- To support bulk purchasing with discounts on bulk orders.
- To allow customers to give product reviews and ratings.

❖ **ABSTRACT**

- Our project will build a simple and user-friendly cloud-based e-commerce website where customers can browse products, add items to the cart, and place orders easily.
- It provides a seller dashboard to manage products and orders efficiently using cloud storage and services.
- The website uses cloud technology for faster performance, secure data handling, and the ability to handle more users as the store grows.

❖ LITERATURE REVIEW

1. **Amazon & Flipkart:** Popular e-commerce platforms that use cloud technology for fast performance, product search, secure payments, and large-scale user handling.
2. **Shopify:** A cloud-based e-commerce system offering easy product management and secure checkout.
3. **Cloud Services (AWS / Google Cloud):** Used by modern e-commerce websites for hosting, storage, security, and scalability.

Amazon and Flipkart mainly focus on large sellers, making them complex and expensive for small and local businesses.

This project proposes a simple, affordable, and scalable cloud-based e-commerce website designed for small and local sellers, offering features like product listing, secure payment, and order management.

The system also supports **bulk product purchases with discounts**, enabling wholesale buying and improving business opportunities for sellers.

❖ USED TECH STACK

- **Frontend:** React.js, CSS
- **Backend:** Node.js, Express.js
- **Database:** MongoDB

MongoDB - to store product, user, and order data

Express.js - to manage server-side routes and APIs

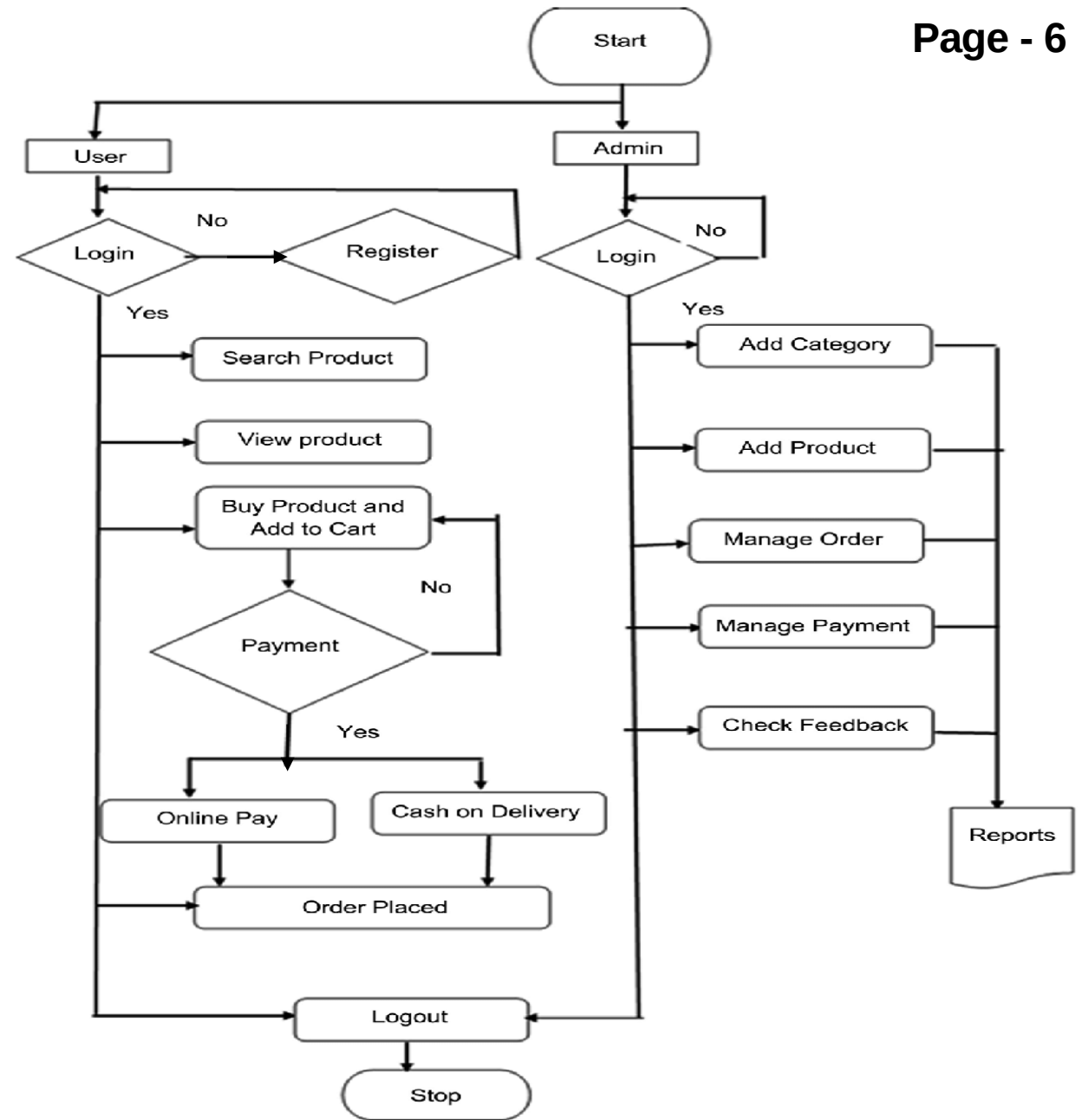
React.js - to build a responsive frontend interface

Node.js - as the backend runtime environment



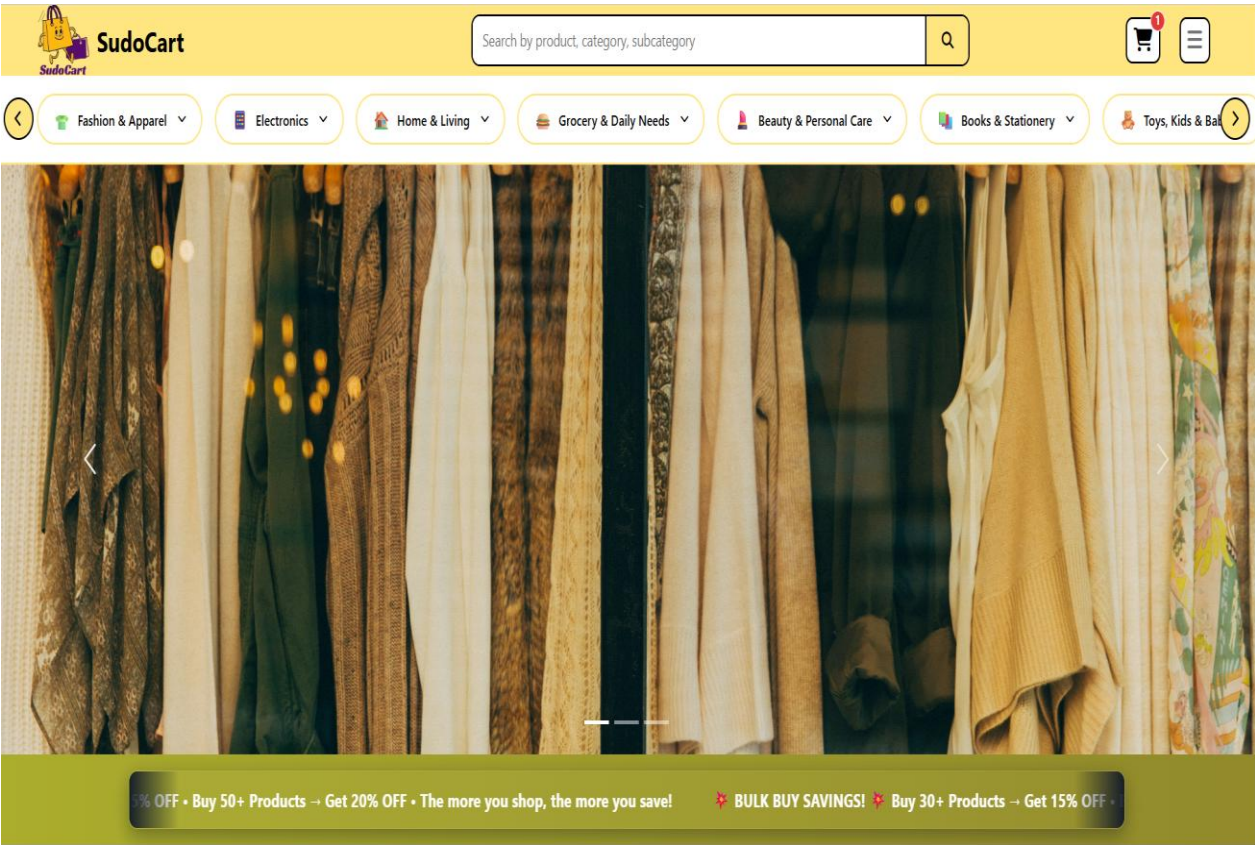
PROPOSED APPROACH

The proposed approach focuses on building a user-friendly, secure, and scalable cloud-based e-commerce website. It ensures real-time product updates, smooth shopping, and efficient admin management. Post-launch, feedback will guide continuous improvements and feature enhancements.

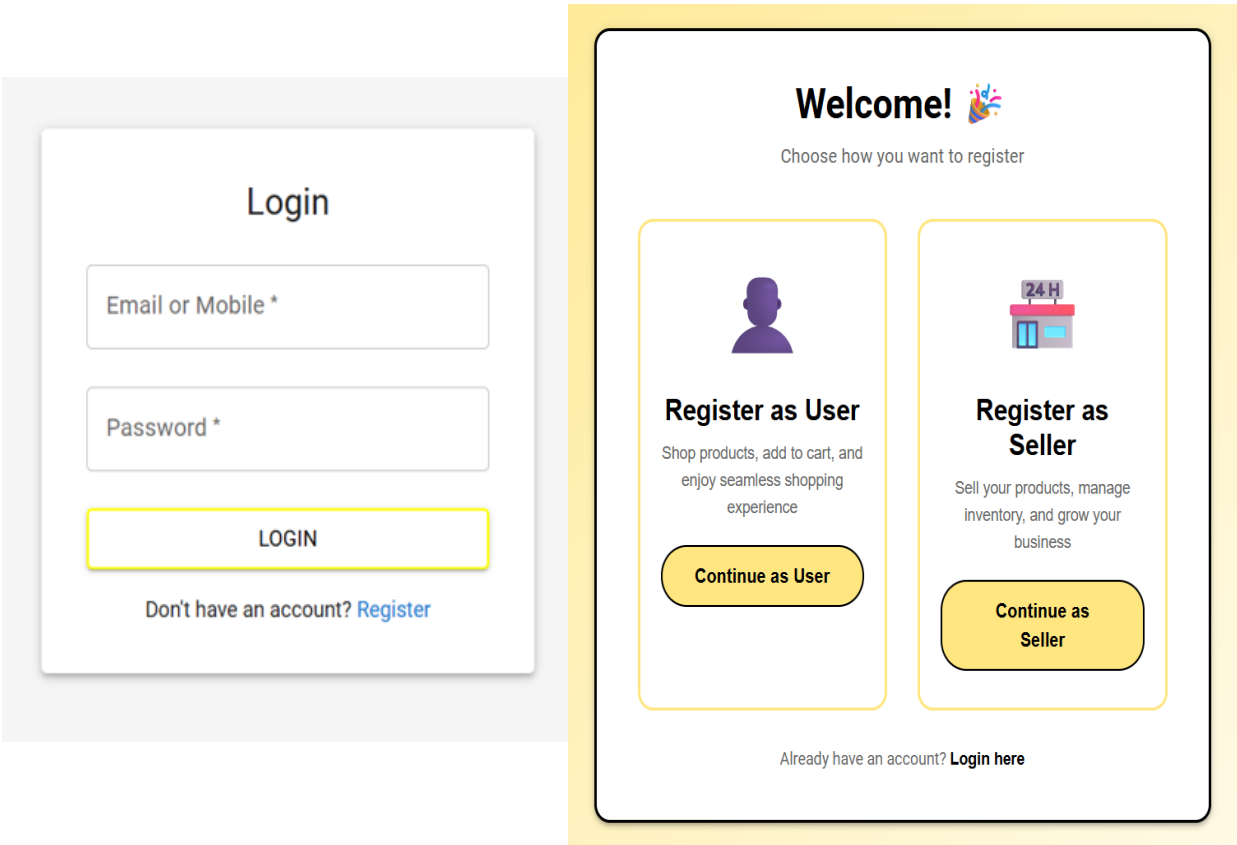


OVERVIEW OF USER INTERFACE

Home Page



Login & Sign up Page



OVERVIEW OF USER INTERFACE

Featured Categories

All Categories



Samsung galaxy
₹18999 ₹29999 37%
[Show more](#)



Laptop
₹49999 ₹69999 29%
[Show more](#)



Smart Watch
₹1999 ₹2299 13%
[Show more](#)



Tablet
₹39999 ₹49999 20%
[Show more](#)



T-Shirt
₹599 ₹997 40%
[Show more](#)



shirt
₹999 ₹1298 0%
[Show more](#)



Hoodies
₹1699 ₹2298 26%
[Show more](#)



jeans
₹1 ₹997 100%
[Show more](#)

**SudoCart**

Search product by name

Login





Home Shop Men Women Kids About

Featured Categories

[SHOP NOW](#)

[SHOP NOW](#)

[SHOP NOW](#)

Featured Items [Show all →](#)

**SudoCart**

Search product by name

Login





Home Shop Men Women Kids About



T-Shirt
₹210 ₹599 65% off
[Show more](#)



Shirt
₹390 ₹599 35% off
[Show more](#)



Frocks
₹650 ₹999 35% off
[Show more](#)



Gown
₹750 ₹1499 50% off
[Show more](#)

SudoCart


Contact Us
91+ 9523599608
sudocart.com
Buxar, Bihar

Important Links
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OVERVIEW OF USER INTERFACE

Cart Page

Your Cart

Product	Price	Quantity	Delete
 Samsung galaxy	₹1263	+ 5 -	
 T-Shirt 15% Bulk Discount	₹599 ₹509.15	+ 30 -	
 shirt 20% Bulk Discount	₹999 ₹799.20	+ 51 -	

 Bulk Discount Info:

• Buy 30-49 of the same product: Get 15% discount

• Buy 50+ of the same product: Get 20% discount

Proceed to Pay

Total: ₹62348.70

Watchlist Page

Your Watchlist

 T-Shirt
₹249

move to cart

 Formal Shirt
₹300

move to cart

 Kurtis
₹200

move to cart

OVERVIEW OF USER INTERFACE

Product Details



T-Shirt

Premium cotton shirt with a perfect fit, soft feel, and modern look. Ideal for daily wear and semi-formal occasions.

₹599

₹997

40%

Category: Fashion & Apparel

Subcategory: Men Clothing

Size:

M

Add to Cart

Buy Now

Customer Reviews

4.0

1 reviews

5 stars

4 stars

3 stars

2 stars

1 star

0

1

0

0

0

Amit Kumar

March 9, 2026

★★★★★

This product quality is very Good.

Checkout

Checkout & Payment

CUSTOMER INFO

Name: Amit Kumar

Email: kumaramitbxr2005@gmail.com

Mobile: 1234567890

Delivery Address

☒ Default Registered Address

Bbit Boys Hostel, Kolkata, West Bengal - 700137

☐ Use Custom / New Address

Shipping To: Bbit Boys Hostel, Kolkata, West Bengal - 700137

Items in Your Order

 Kurti

₹399.00 × 1

₹399.00

Total Payable Amount

₹399.00

Select Payment Method

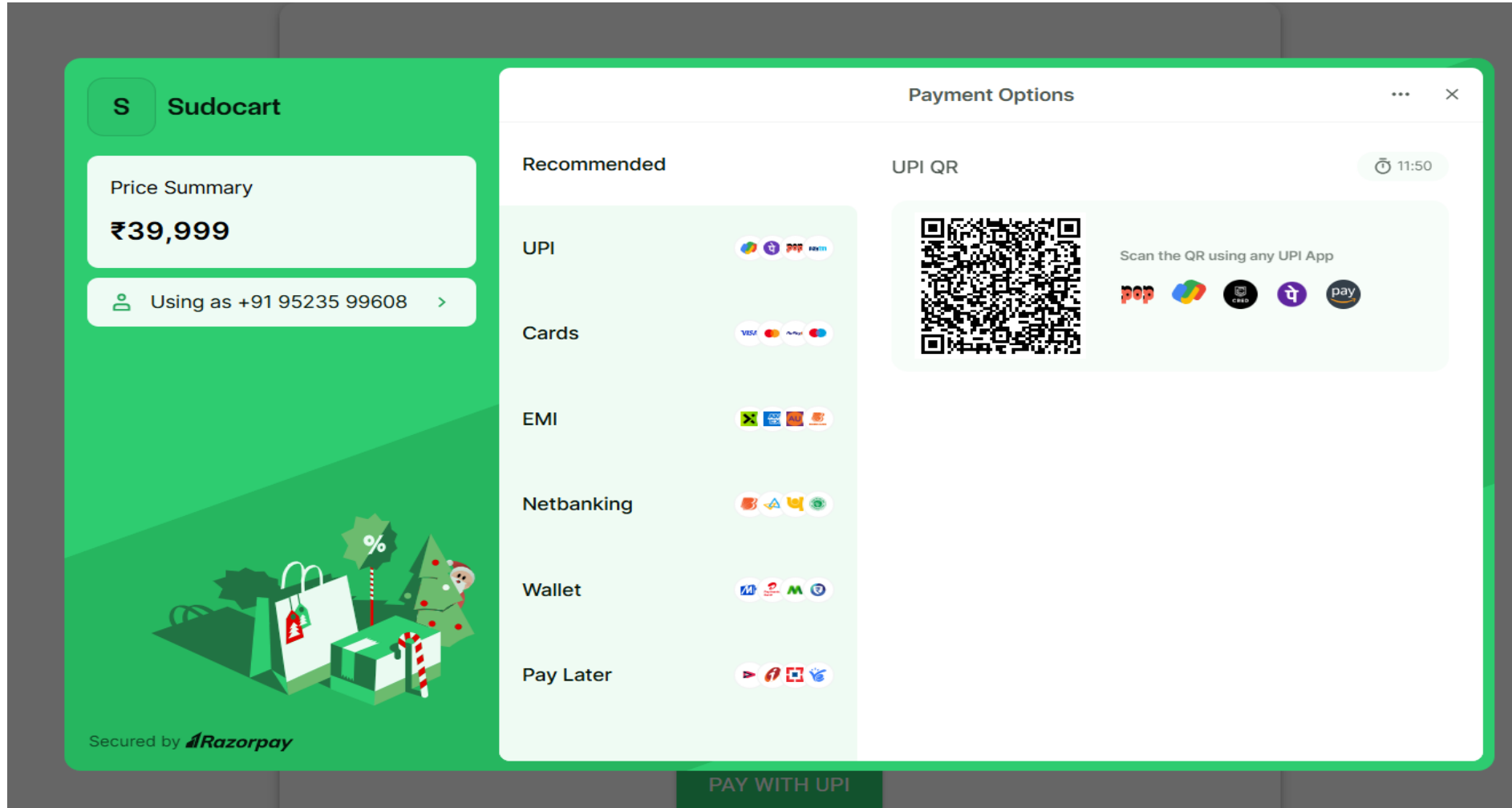
☒ Cash on Delivery (COD)

☐ Online Pay (UPI/Cards/Netbanking)

Confirm & Place Order

OVERVIEW OF USER INTERFACE

Online Payment & Order Placed



OVERVIEW OF SELLER INTERFACE

Seller Dashboard & Product management

Seller Dashboard

Add New Product

Amit Kumar

Logout

Product Analytics

Total Products

8

Total Stock

223

Total Value

₹35,21,599

Order Analytics

Total Orders

7

Ordered

4

Dispatched

1

Delivered

2

Total Revenue

₹99,595

My Products

Orders



Samsung galaxy

Electronics

₹18999 ₹20999

37%

Stock: 94

Edit

Delete



Laptop

Electronics

₹49999 ₹60999

29%

Stock: 24

Edit

Delete



Smart Watch

Electronics

₹1999 ₹2299

13%

Stock: 8

Edit

Delete



Tablet

Electronics

₹39999 ₹49998

20%

Stock: 11

Edit

Delete



Welcome, Amit Kumar



Personal Information

Name: Amit Kumar

Email: kumaramitbxr2004@gmail.com

Mobile: 9523599608

Role: Seller



Business Information

Shop/Brand Name: Electronics wala

Business Category: Electronics

Description: I am 10 years in this business.



Legal Documents

PAN Card: ASDF1234

Aadhaar: 1234512345

GST Number: QWERT12345



Business Address

Address: Kharahana

City: Buxar

State: Bihar

Pincode: 802123



Return Address

Address: KHARAHANA

City: Buxar

State: Bihar

Pincode: 802123

LOGOUT



Edit Product

Update your product details



Change Image

OR

Image URL

data:image/png;base64,/9j/AAQSkZJRgABAAQ

Product Information

*Product Name *

Samsung galaxy

*Main Category *

Electronics

*Sub Category *

Mobiles & Smartphones

*Description

This is a good product.

Pricing Details

Selling Price *

₹ 18999

Original Price *

₹ 25999

Stock Quantity *

94

*Calculated Discount

37% OFF

Auto-calculated based on prices

Cancel

Update Product

OVERVIEW OF ADMIN INTERFACE

Admin Login & Order Management

SudoCart

Admin Panel Login

Authorized admins only

Username *

Password *

Login to Dashboard

Admin Dashboard

Manage orders, users, and sellers from one place.

Total Orders

1

Users

2

Sellers

1

Today's Logins

2

Orders

Users & Sellers

Login Details (Users + Sellers)

Name	Email	Mobile	Role	Last Login	Login Count	Action
MANISH KUMAR	nitishkumarbuxar802103@gmail.com	9142019687	User	6/7/2026, 8:41:07 PM	1	View
Amit Kumar	kumaramitbxr2004@gmail.com	9523599608	Seller	6/15/2026, 1:11:46 AM	38	View
Amit Kumar	kumaramitbxr2005@gmail.com	1234567890	User	6/15/2026, 1:33:15 AM	39	View

Contact Us

Say Hello.

We'd love to hear from you — whether you have a question about a product, feedback to improve our service, or just want to say hello. Our support team is here for you — fast, friendly, and always ready to assist. Reach out via email, call, or follow us on social media.

At ShopKart, your satisfaction is our top priority. Let's stay connected!

Buxar Bihar India

sudocart@gmail.com

91+ 9523599608



Ask Your Queries

Name

Email

Phone

Message

Submit

OVERVIEW OF BACKEND SURFACE

Database

MongoDB Compass - mongoDBNew/sudocart1.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

chatbot

mongoDBNew

Mongodb_proj_database

admin

chatbot

config

enqueryDB

jharkhand_tourism

local

myClgDB

quiz

salon

sudocart1

carts

orders

products

users

watchlists

SalonDB

ordersuserscarts+

mongoDBNew > sudocart1 > users

Documents (3)AggregationsSchemaIndexes (1)Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

UPDATE

DELETE

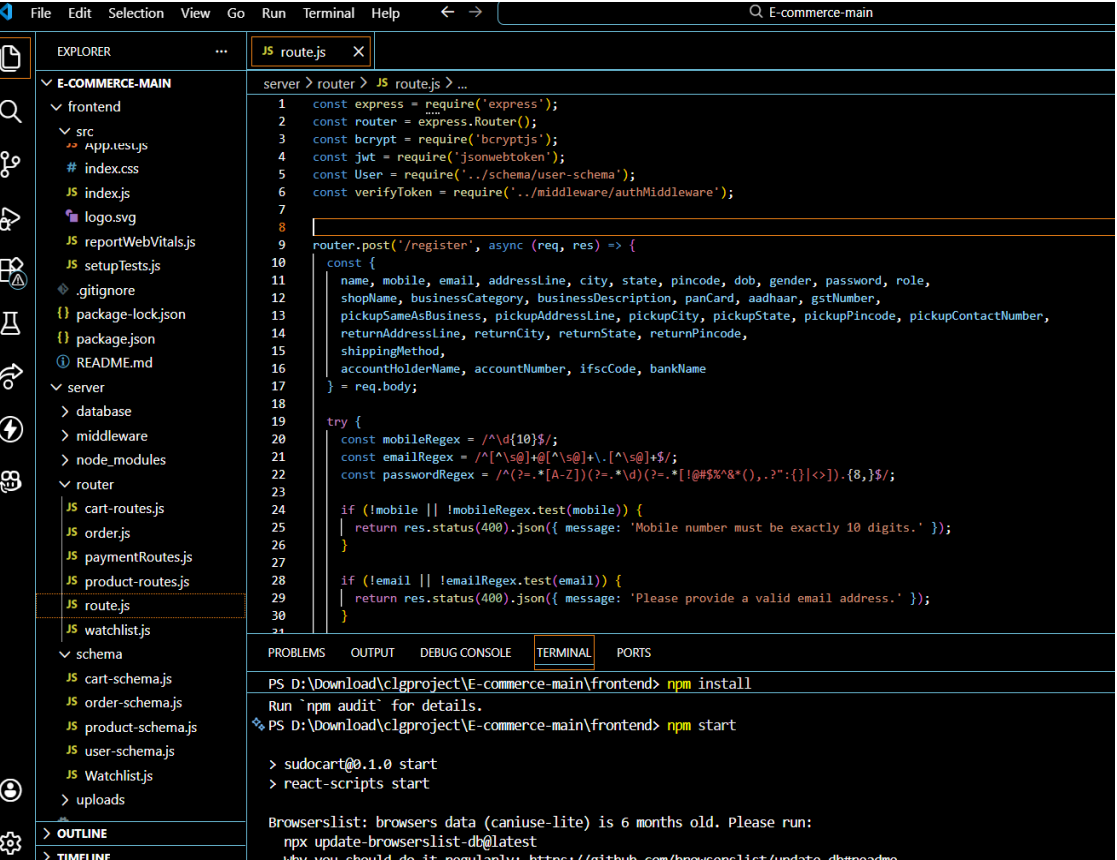
75

```
_id: ObjectId('693eb5e1e693963b9ce6c8ed')
name: "Amit "
email: "kumaramitbxr2005@gmail.com"
mobile: "1234567890"
password: "$2b$10$KkzJqAZ7YsYo2tZPZ9HkCudeMBHoSfLnkFNA1snbD1cfjqxvqlVWC"
role: "user"
addressLine: "Bbit Boys Hostel"
city: "Kolkata"
state: "West Bengal"
pincode: "700137"
dob: "2005-11-02"
gender: "male"
pickupSameAsBusiness: "yes"
shippingMethod: "self"
createdAt: 2025-12-14T13:04:33.584+00:00
updatedAt: 2025-12-14T13:04:33.586+00:00
__v: 0
```

```
_id: ObjectId('693ebb44e693963b9ce6c905')
name: "Amit Kumar"
email: "kumaramitbxr2004@gmail.com"
mobile: "9523599608"
password: "$2b$10$4fKRtDwnldVJGBazinY0muNLCU84C824A0VKvuWw3p9uEcqdfnLAY"
role: "seller"
addressLine: "Kharahana"
city: "Buxar"
state: "Bihar"
```

OVERVIEW OF BACKEND SURFACE

Login & Register Page logic Backend code



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the `server > router > JS routes.js` file. The code defines a `router.post('/register')` endpoint. It includes validation for mobile number and email. The terminal at the bottom shows the command `npm install` and `npm start` being executed.

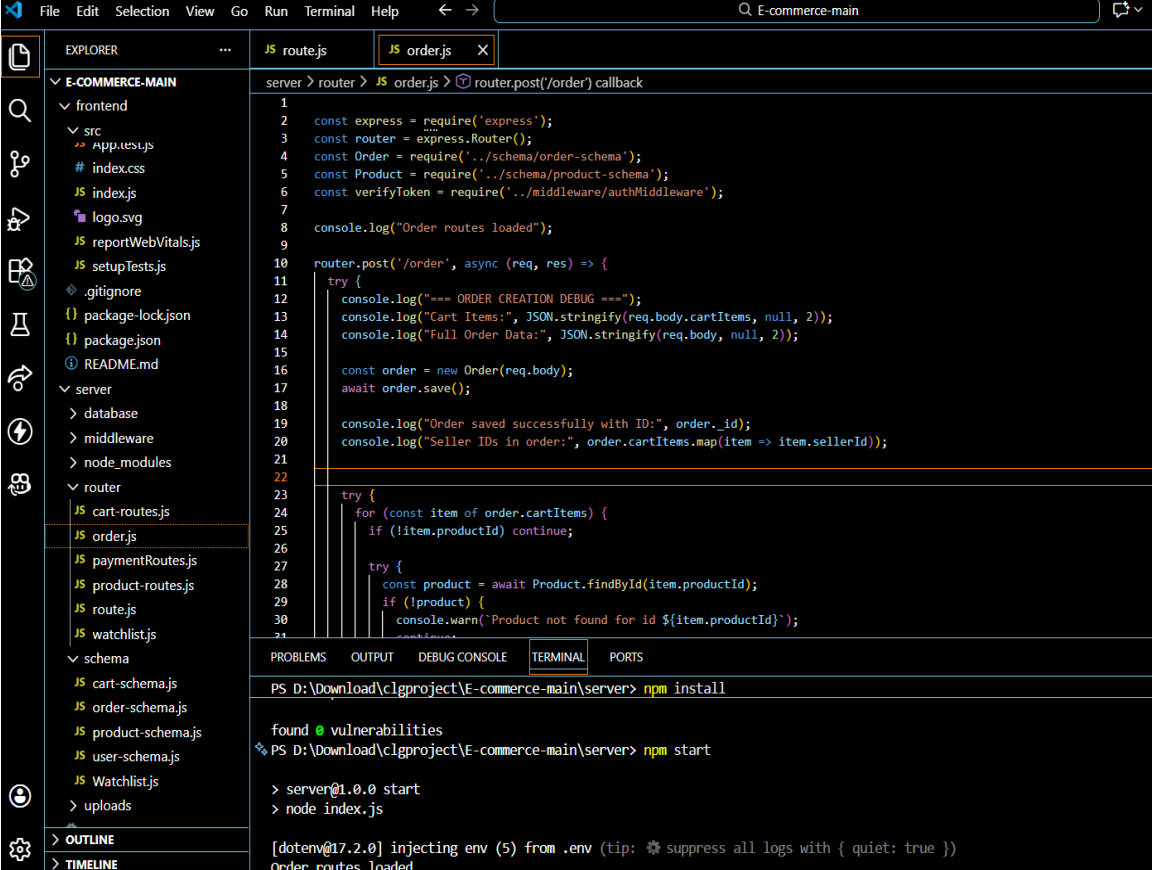
```
1 const express = require('express');
2 const router = express.Router();
3 const bcrypt = require('bcryptjs');
4 const jwt = require('jsonwebtoken');
5 const User = require('../schema/user-schema');
6 const verifyToken = require('../middleware/authMiddleware');
7
8
9 router.post('/register', async (req, res) => {
10   const {
11     name, mobile, email, addressLine, city, state, pincode, dob, gender, password, role,
12     shopName, businessCategory, businessDescription, panCard, aadhaar, gstNumber,
13     pickupSameAsBusiness, pickupAddressLine, pickupCity, pickupState, pickupPincode, pickupContactNumber,
14     returnAddressLine, returnCity, returnState, returnPincode,
15     shippingMethod,
16     accountHolderName, accountNumber, ifscCode, bankName
17   } = req.body;
18
19   try {
20     const mobileRegex = /^[0-9]{10}$/;
21     const emailRegex = /^[^@]+@[^\s@]+\.[^\s@]+$/;
22     const passwordRegex = /^(?=.*[A-Z])(?=.*[a-z])(?=.*[0-9])(?=.*[!@#$%^&*()_~`{}|;':"<>]).{8,}$/;
23
24     if (!mobile || !mobileRegex.test(mobile)) {
25       return res.status(400).json({ message: 'Mobile number must be exactly 10 digits.' });
26     }
27
28     if (!email || !emailRegex.test(email)) {
29       return res.status(400).json({ message: 'Please provide a valid email address.' });
30     }
31   }
32 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Download\clgproject\E-commerce-main\frontend> npm install
Run `npm audit` for details.
PS D:\Download\clgproject\E-commerce-main\frontend> npm start
> sudocart@0.1.0 start
> react-scripts start

Browserslist: browsers data (caniuse-lite) is 6 months old. Please run:
  npx update-browserslist-db@latest
  why you should do it regularly: https://github.com/browserslist/update-db#readme
```

Order Management logic



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the `server > router > JS order.js` file. The code defines a `router.post('/order')` endpoint. It includes validation for the order data and saves the order to the database. The terminal at the bottom shows the command `npm install` and `npm start` being executed.

```
1
2 const express = require('express');
3 const router = express.Router();
4 const Order = require('../schema/order-schema');
5 const Product = require('../schema/product-schema');
6 const verifyToken = require('../middleware/authMiddleware');
7
8 console.log("Order routes loaded");
9
10 router.post('/order', async (req, res) => {
11   try {
12     console.log("=== ORDER CREATION DEBUG ===");
13     console.log("Cart Items:", JSON.stringify(req.body.cartItems, null, 2));
14     console.log("Full Order Data:", JSON.stringify(req.body, null, 2));
15
16     const order = new Order(req.body);
17     await order.save();
18
19     console.log("Order saved successfully with ID:", order._id);
20     console.log("Seller IDs in order:", order.cartItems.map(item => item.sellerId));
21
22   } catch (error) {
23     console.log("Error in order creation:", error);
24   }
25 }
26
27 for (const item of order.cartItems) {
28   if (!item.productId) continue;
29   try {
30     const product = await Product.findById(item.productId);
31     if (!product) {
32       console.warn("Product not found for id ${item.productId}");
33     }
34   }
35 }
```

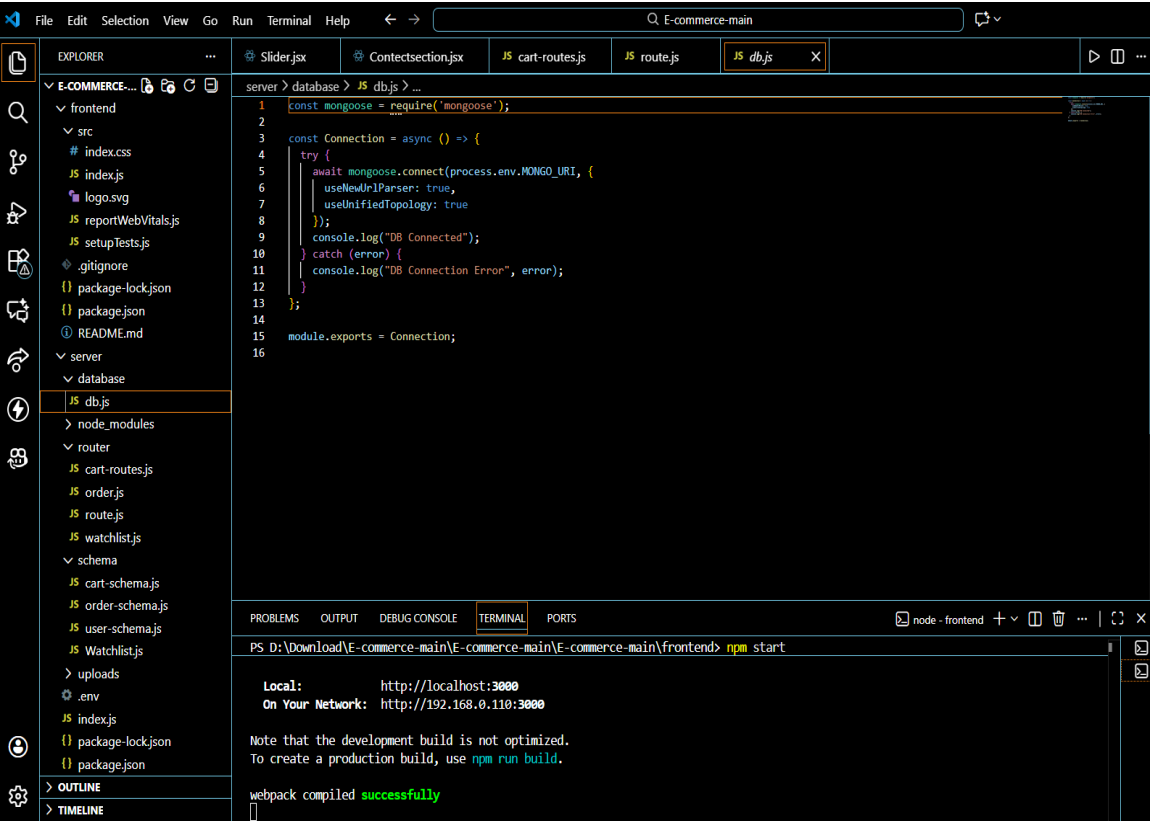
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Download\clgproject\E-commerce-main\server> npm install
found 0 vulnerabilities
PS D:\Download\clgproject\E-commerce-main\server> npm start
> server@1.0.0 start
> node index.js

[dotenv@17.2.0] injecting env (5) from .env (tip: ✨ suppress all logs with { quiet: true })
Order routes loaded
```


OVERVIEW OF BACKEND SURFACE

Database connection logic



The screenshot shows the VS Code interface with the Explorer on the left, the Editor in the center, and the Terminal at the bottom. The Explorer shows the project structure with 'db.js' selected under the 'server' directory. The Editor displays the code for 'db.js', which uses the 'mongoose' library to connect to a MongoDB database. The code includes a 'const mongoose = require('mongoose');' statement, a 'const Connection = async () => {' function, and a 'try' block that calls 'await mongoose.connect(process.env.MONGO_URI, { useNewUrlParser: true, useUnifiedTopology: true });'. It also includes a 'catch' block for errors and a 'module.exports = Connection;' statement. The Terminal shows the command 'npm start' being executed, and the output indicates that the server is running on 'http://localhost:3000' and 'http://192.168.0.110:3000'. A message states: 'Note that the development build is not optimized. To create a production build, use npm run build.' The status bar at the bottom shows 'webpack compiled successfully'.

```
1 const mongoose = require("mongoose");
2
3
4 const Connection = async () => {
5   try {
6     await mongoose.connect(process.env.MONGO_URI, {
7       useNewUrlParser: true,
8       useUnifiedTopology: true
9     });
10    console.log("DB Connected");
11  } catch (error) {
12    console.log("DB Connection Error", error);
13  }
14 }
15
16 module.exports = Connection;
```

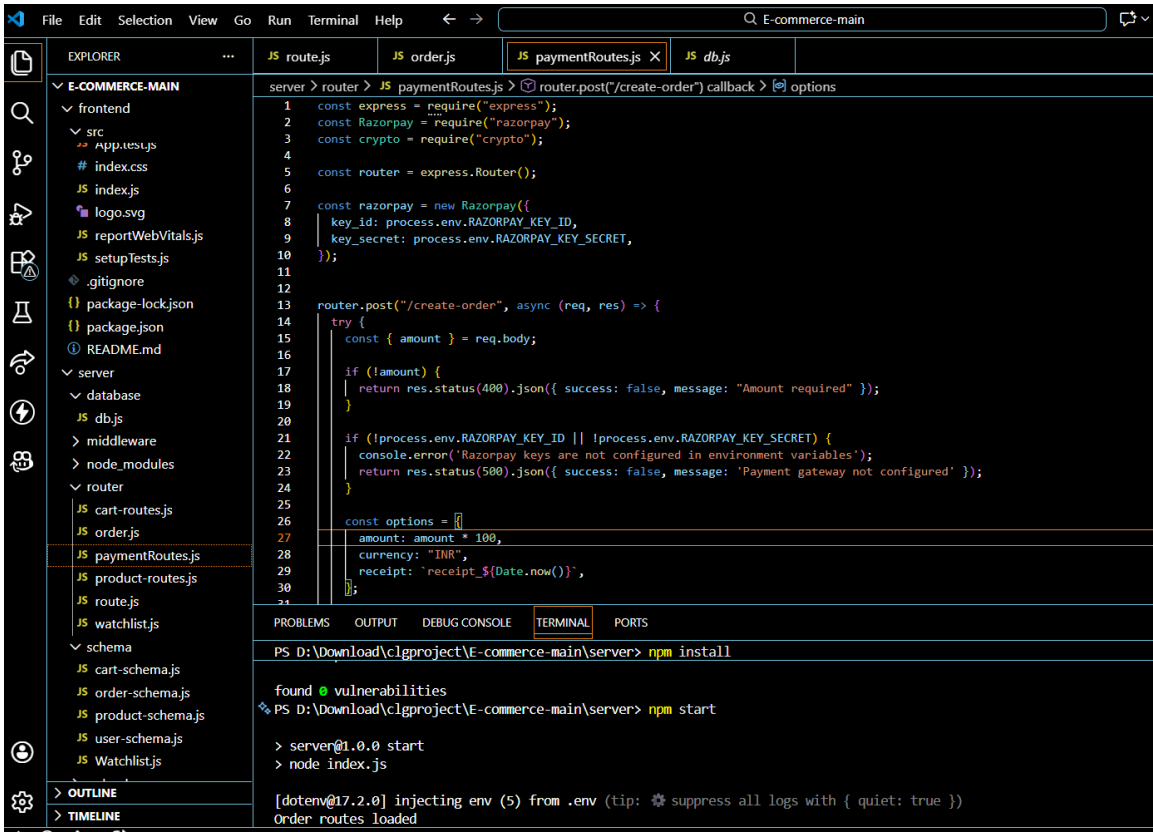
PS D:\Download\E-commerce-main\E-commerce-main\frontend> npm start

Local: http://localhost:3000
On Your Network: http://192.168.0.110:3000

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled successfully

Payment logic Backend code



The screenshot shows the VS Code interface with the Explorer on the left, the Editor in the center, and the Terminal at the bottom. The Explorer shows the project structure with 'paymentRoutes.js' selected under the 'server' directory. The Editor displays the code for 'paymentRoutes.js', which uses 'express' and 'razorpay' to handle payment requests. The code includes a 'const express = require('express');' statement, a 'const razorpay = new Razorpay({ key_id: process.env.RAZORPAY_KEY_ID, key_secret: process.env.RAZORPAY_KEY_SECRET, });' statement, and a 'router.post('/create-order', async (req, res) => {' function. It includes a 'try' block that checks if the amount is provided and if the razorpay keys are configured. It also includes a 'const options = {' statement and a 'return res.status(200).json({ success: true, message: 'Payment gateway not configured' });' statement. The Terminal shows the command 'npm install' being executed, and the output indicates that the server is running on 'http://localhost:3000' and 'http://192.168.0.110:3000'. A message states: 'Note that the development build is not optimized. To create a production build, use npm run build.' The status bar at the bottom shows 'webpack compiled successfully'.

```
1 const express = require("express");
2 const Razorpay = require("razorpay");
3 const crypto = require("crypto");
4
5 const router = express.Router();
6
7 const razorpay = new Razorpay({
8   key_id: process.env.RAZORPAY_KEY_ID,
9   key_secret: process.env.RAZORPAY_KEY_SECRET,
10 });
11
12
13 router.post("/create-order", async (req, res) => {
14   try {
15     const { amount } = req.body;
16
17     if (!amount) {
18       return res.status(400).json({ success: false, message: "Amount required" });
19     }
20
21     if (!process.env.RAZORPAY_KEY_ID || !process.env.RAZORPAY_KEY_SECRET) {
22       console.error("Razorpay keys are not configured in environment variables");
23       return res.status(500).json({ success: false, message: "Payment gateway not configured" });
24     }
25
26     const options = {
27       amount: amount * 100,
28       currency: "INR",
29       receipt: `receipt_${Date.now()}`,
30     };
31
32     const order = await razorpay.orders.create(options);
33
34     return res.status(200).json({ success: true, message: "Order created successfully", order_id: order.id });
35   } catch (error) {
36     console.error("Error creating order", error);
37     return res.status(500).json({ success: false, message: "Internal server error" });
38   }
39 }
40
41 module.exports = router;
```

PS D:\Download\clgproject\E-commerce-main\server> npm install

found 0 vulnerabilities

PS D:\Download\clgproject\E-commerce-main\server> npm start

> server@1.0.0 start
> node index.js

[dotenv@17.2.0] injecting env (5) from .env (tip: ✨ suppress all logs with { quiet: true })

Order routes loaded

❖ FUTURE WORK

Key areas of future work include:

1. **AI recommendations:** Personalize suggestions using browsing & purchase history.
2. **Android app:** Native app for push notifications, offline browsing, and better performance.
3. **Security:** Implement MFA, token rotation, and role-based access control.
4. **Payments:** Integrate wallets, and international cards for broader reach.



CONCLUSION

Our cloud-based e-commerce website provides an easy, secure, and efficient platform for online shopping. It includes all essential features such as product browsing, cart management, order placement, and an admin dashboard for smooth management. Using cloud technology, the system ensures better performance, data security, and scalability for future growth. This project successfully achieves its goal of creating a modern, reliable, and user-friendly e-commerce solution for both customers and administrators.



Thank You